

Lab Summary Report: IAM & Linux Hardening

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Duration: 45 Minutes

PART 1: IAM Governance (10 pts)

Task 1 — Create Administrative Group (5 pts)

Action:

bash

groupadd sysadmins

usermod -aG sysadmins alice

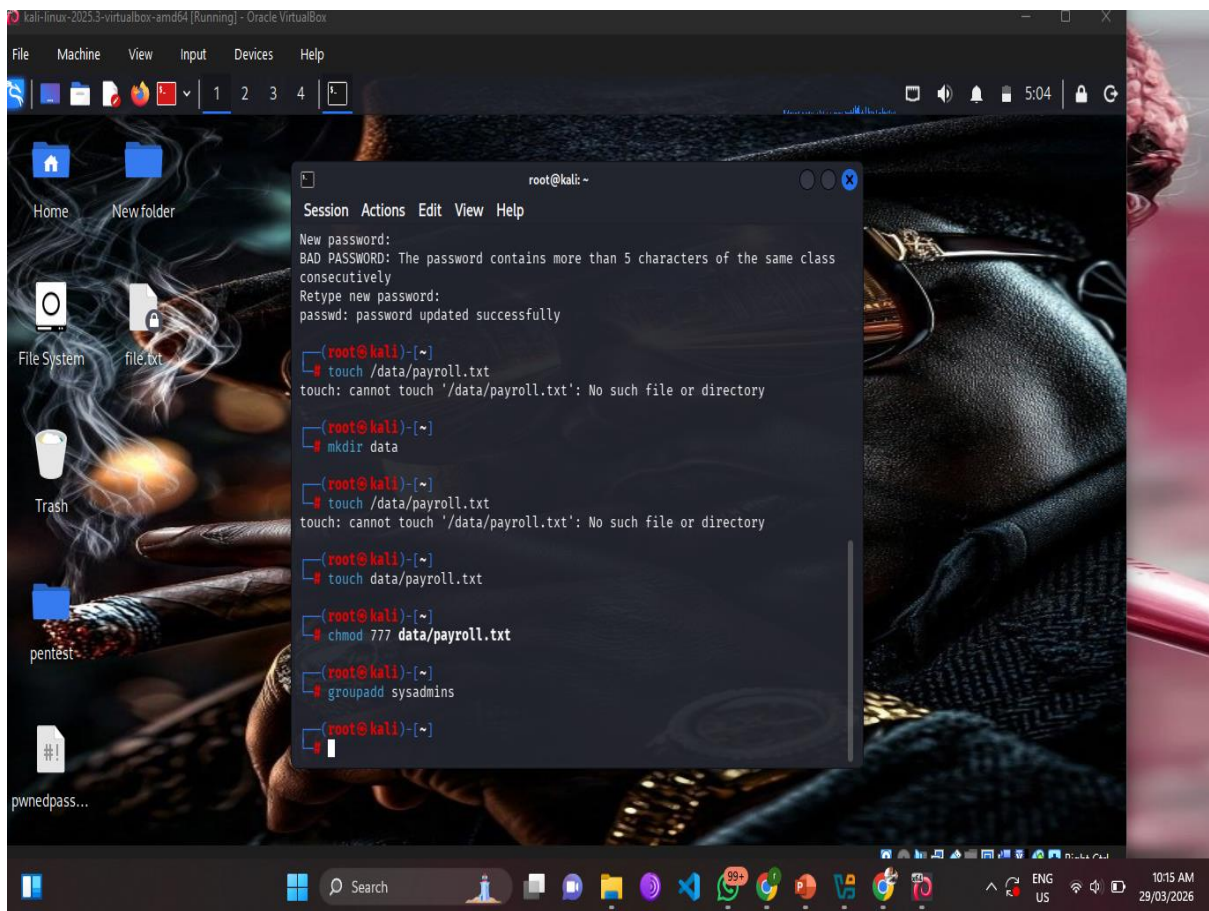
usermod -aG sysadmins bob

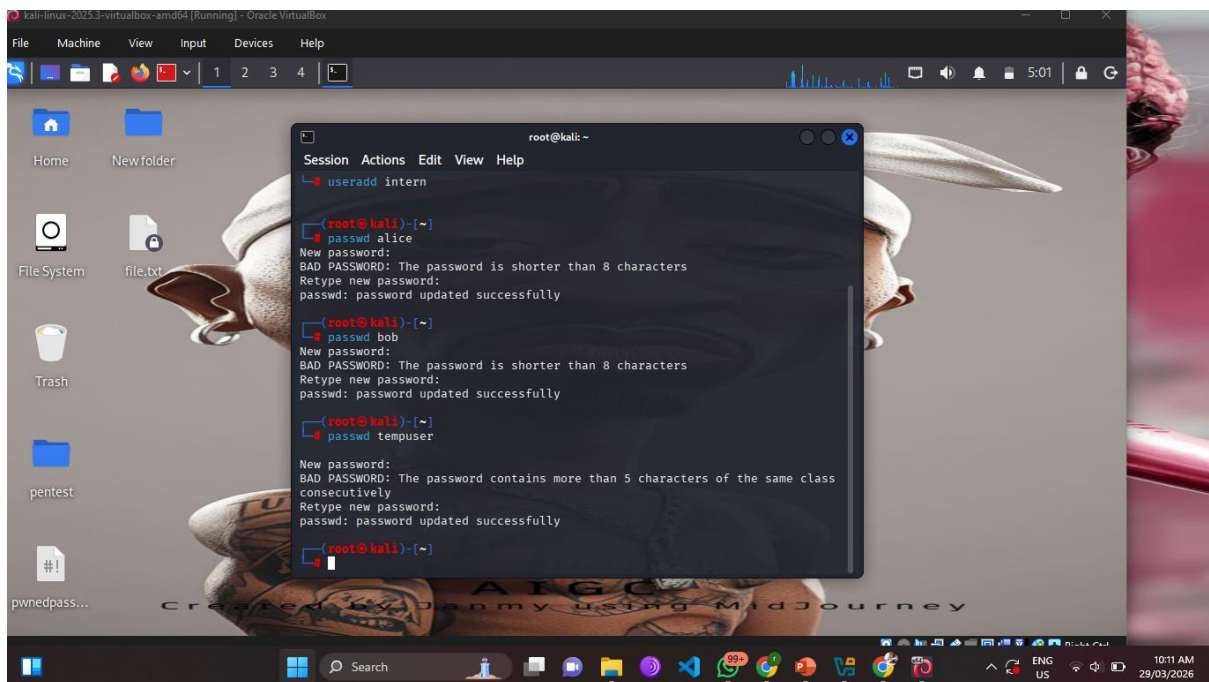
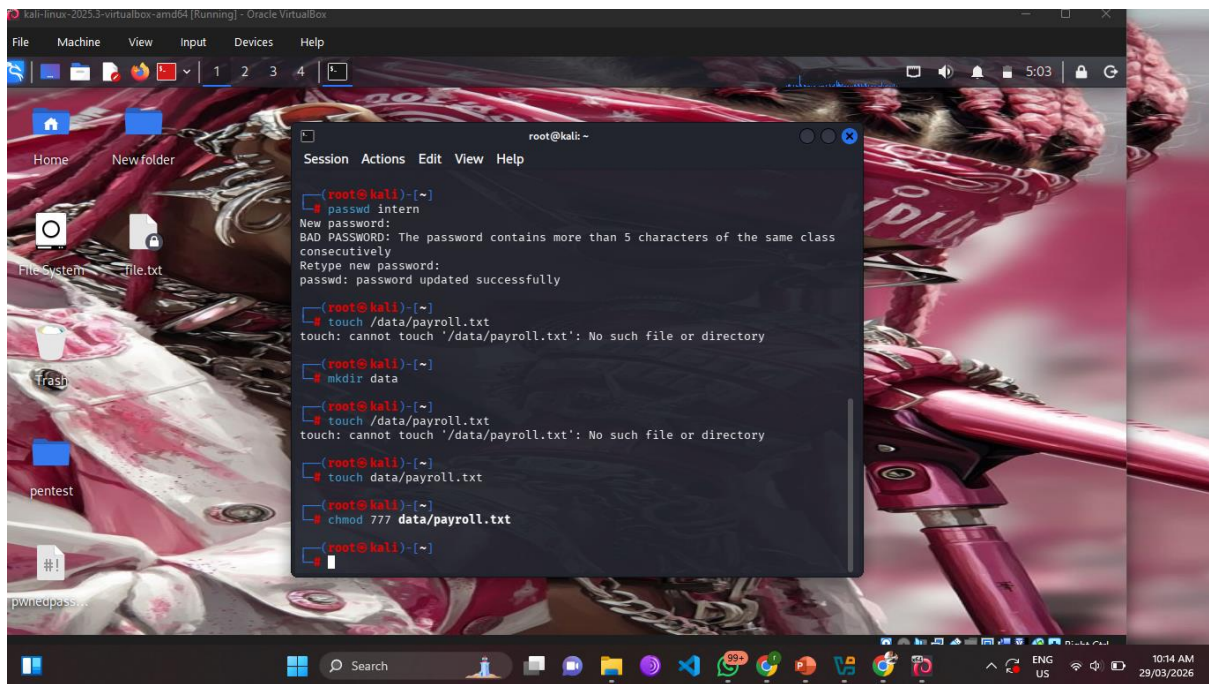
Verification:

bash

getent group sysadmins

Output: sysadmins:x:1002:alice,bob





Task 2 — Configure sudo via visudo (5 pts)

Action:

bash

sudo visudo

Line added: %sysadmins ALL=(ALL) ALL

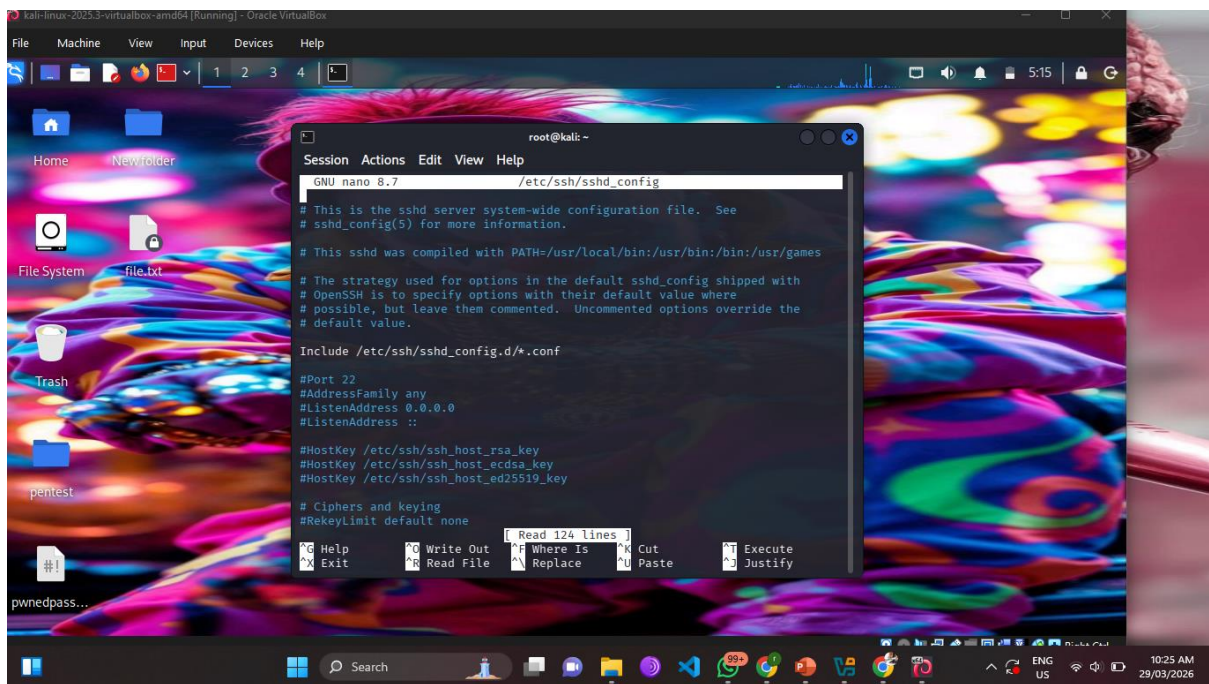
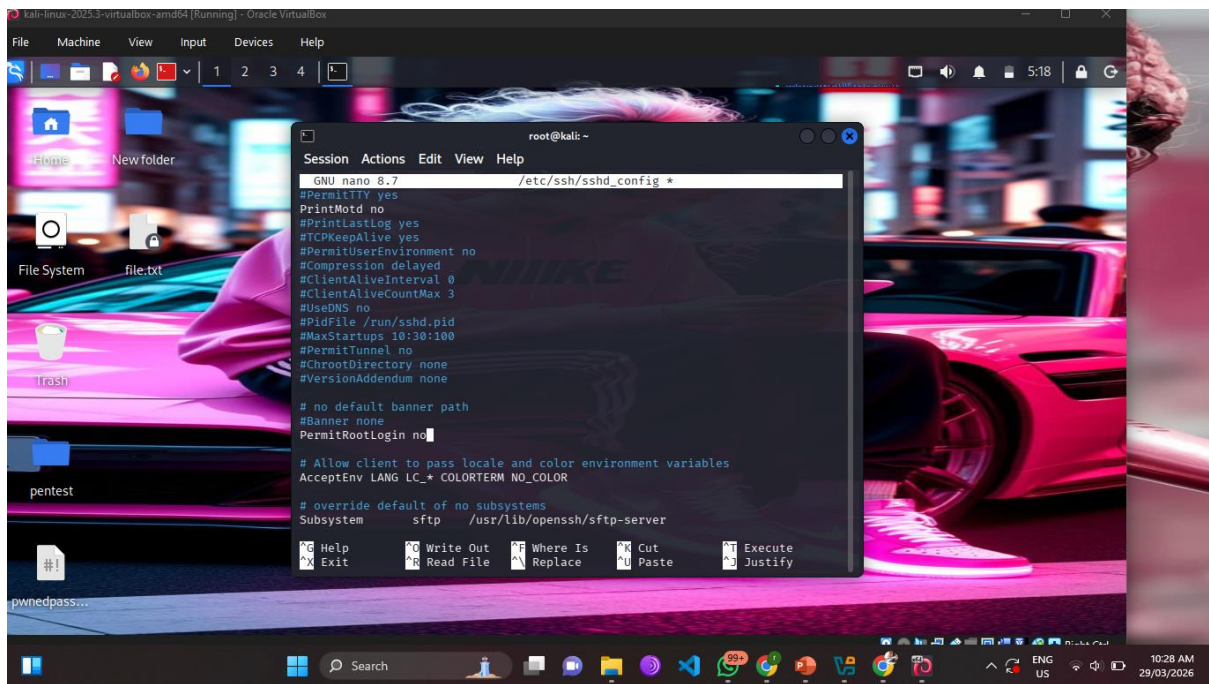
Verification:

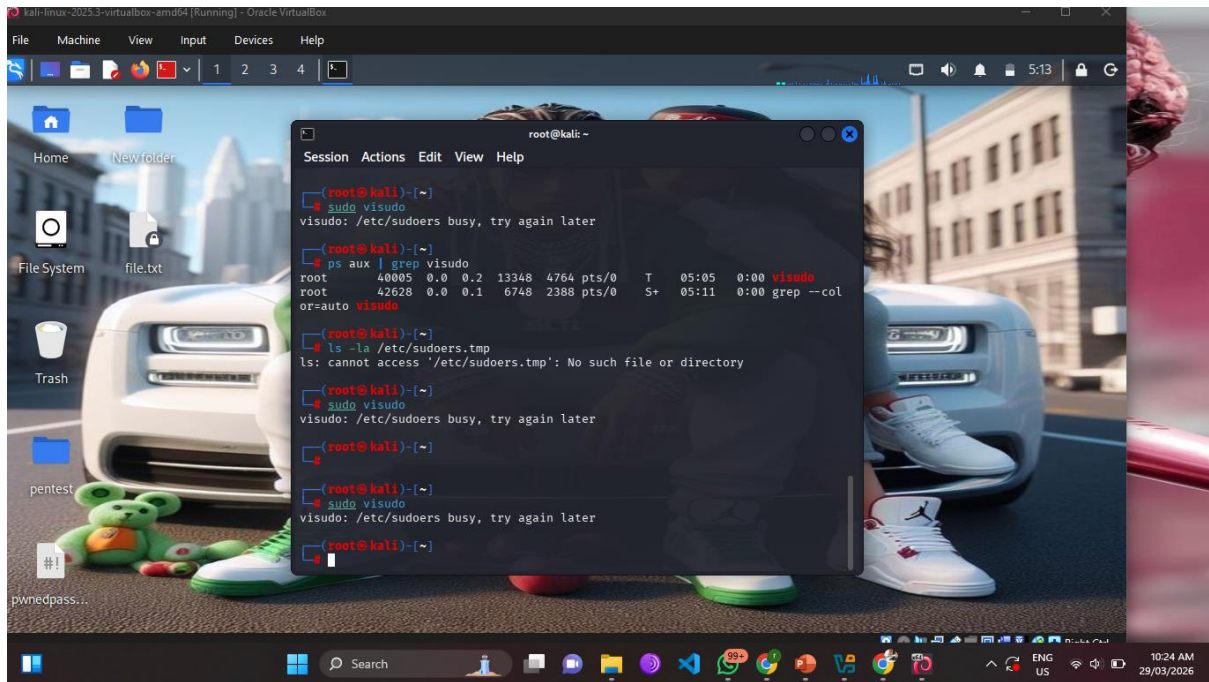
bash

sudo -l -U alice

Output: User alice may run sudo on ALL

[Screenshot Here]





PART 2: Secure Privileged Access (15 pts)

Task 3 — Disable Direct Root Login (5 pts)

Action:

bash

sudo nano /etc/ssh/sshd_config

Change: PermitRootLogin no

Restart:

bash

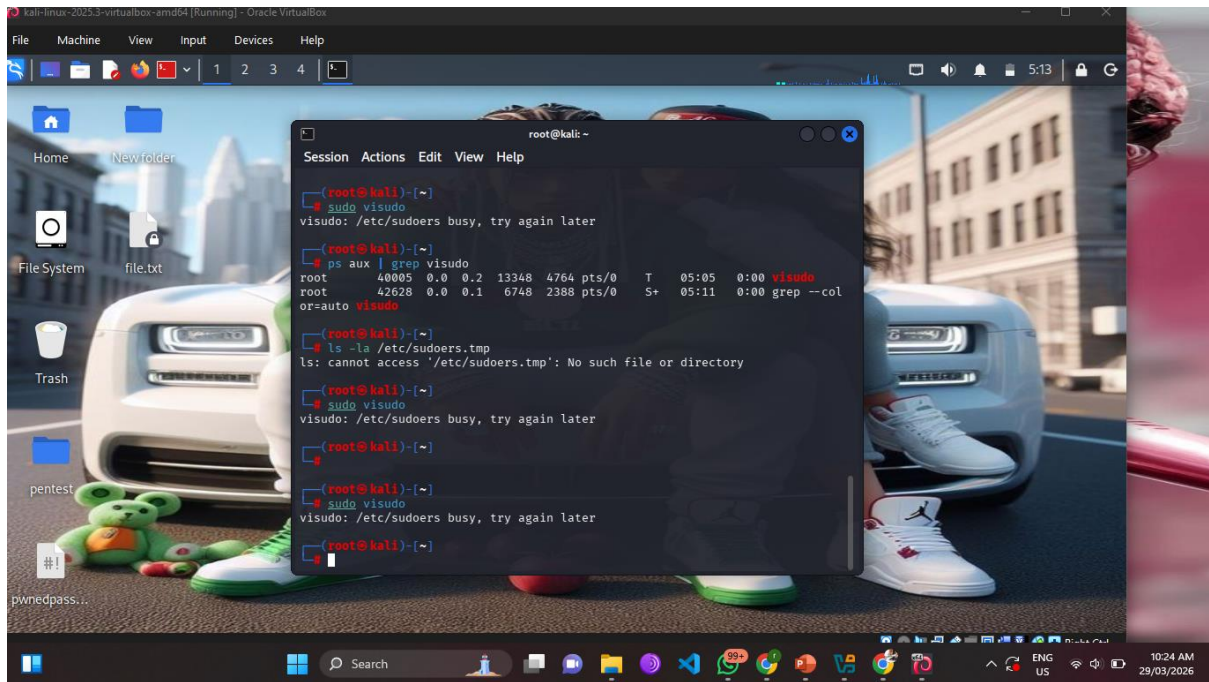
sudo systemctl restart sshd

Verification:

bash

sudo grep PermitRootLogin /etc/ssh/sshd_config

Output: PermitRootLogin no



Task 4 — Configure sudo timeout (5 pts)

Action:

bash

sudo visudo

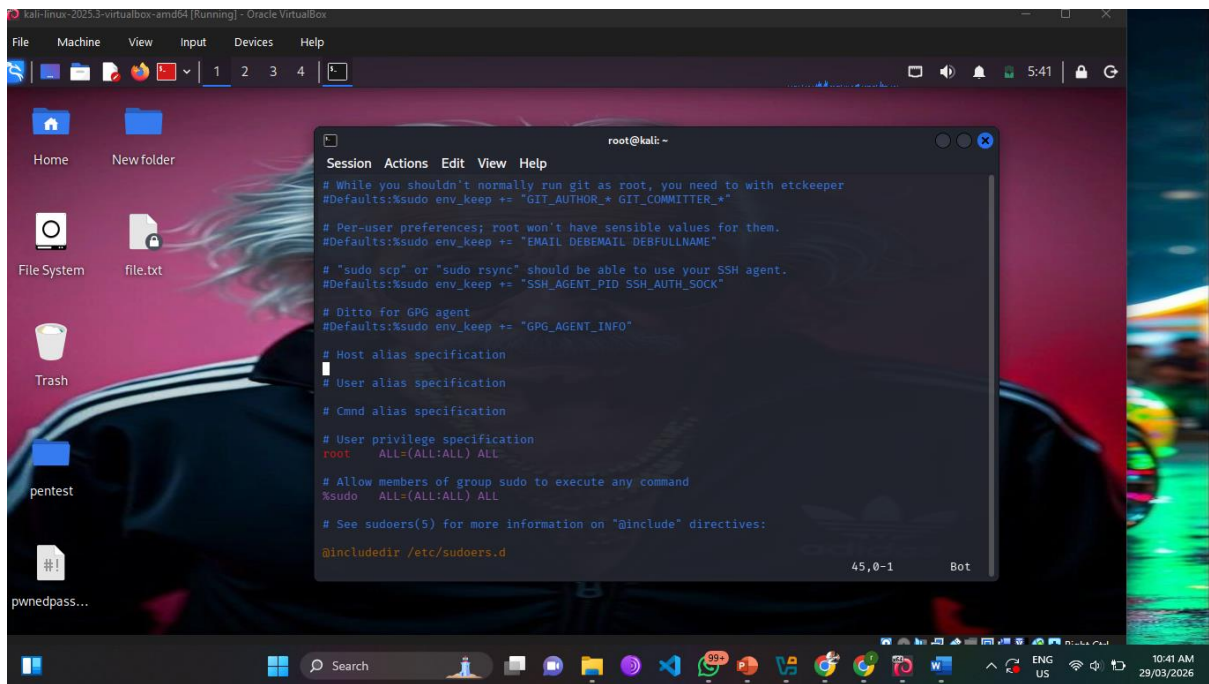
Line added: Defaults timestamp_timeout=3

Verification:

bash

sudo grep timestamp_timeout /etc/sudoers

Output: Defaults timestamp_timeout=3



Task 5 — Create sudo Alias (5 pts)

Action:

bash

sudo visudo

Lines added:

text

User_Alias ADMINS = alice,bob

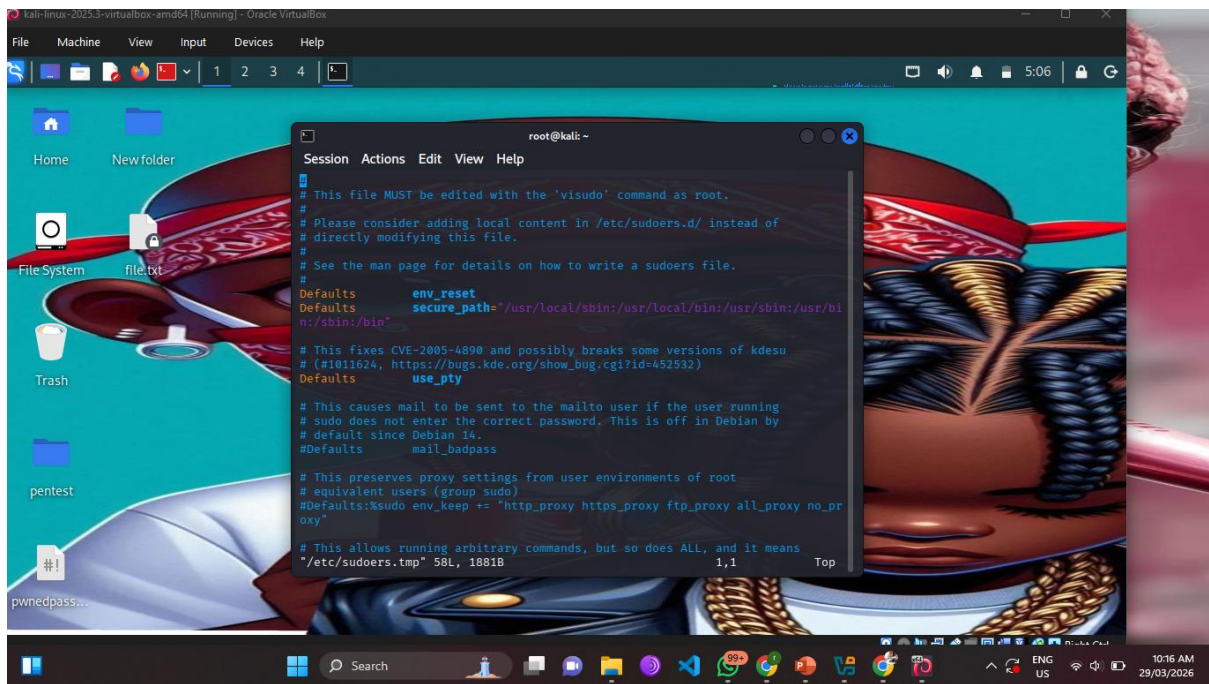
ADMINS ALL=(ALL) ALL

Verification:

bash

sudo -l -U bob

Output: User bob may run sudo on ALL



PART 3: User Hardening (15 pts)

Task 6 — Secure Home Directories (5 pts)

Action:

bash

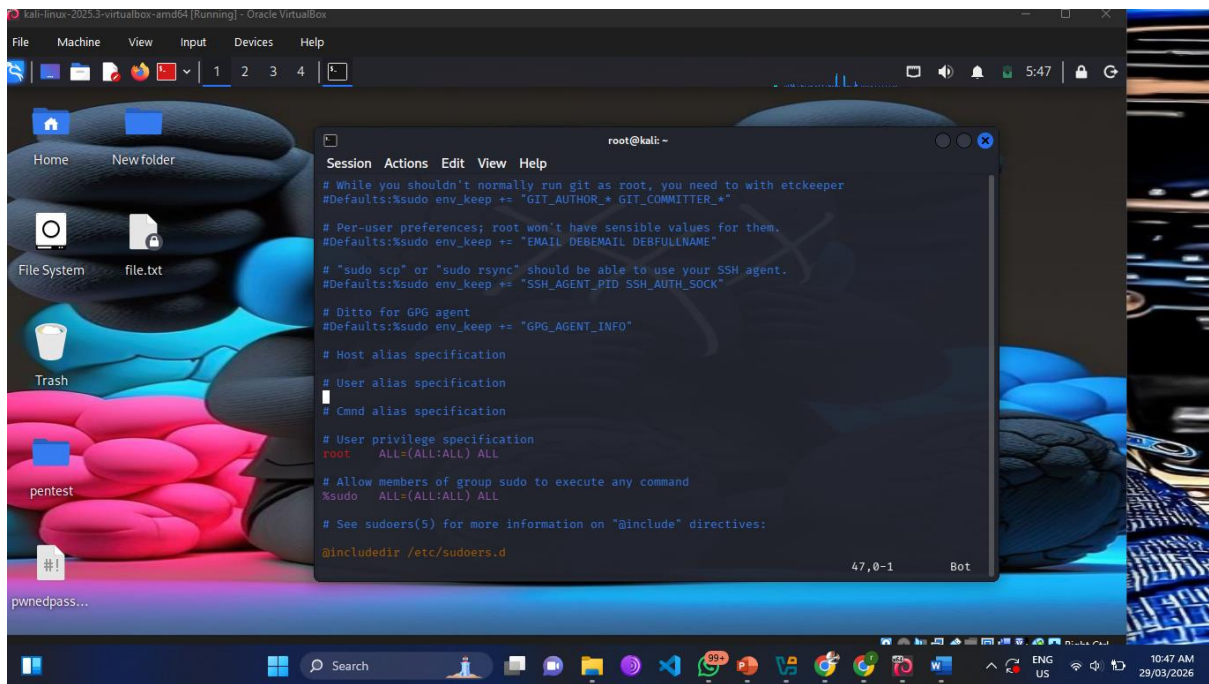
sudo chmod 700 /home/*

Verification:

bash

ls -ld /home/*

Output: drwx----- for each user



Task 7 — Configure Umask (5 pts)

Action:

bash

sudo nano /etc/profile

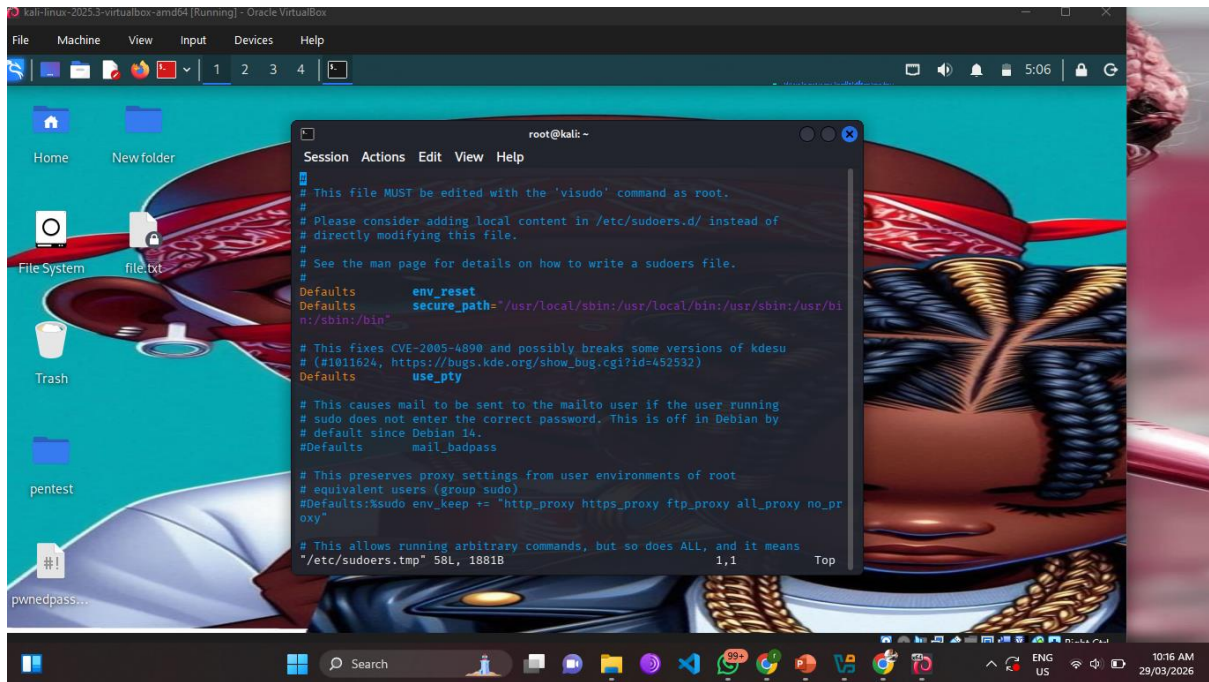
Line added: umask 027

Verification:

bash

umask

Output: 0027



Task 8 — Enforce Strong Password Policy (5 pts)

Action:

bash

sudo nano /etc/security/pwquality.conf

Lines added:

text

minlen = 12

ucredit = -1

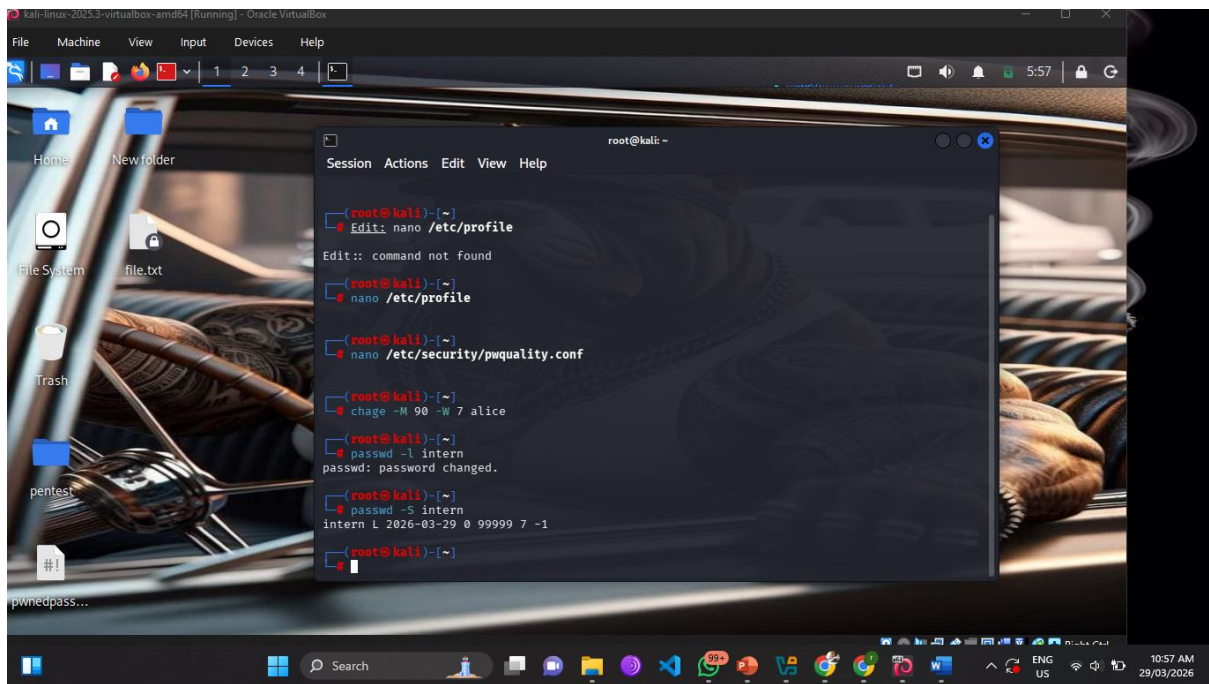
lcredit = -1

dcredit = -1

Verification:

bash

grep -v "^#" /etc/security/pwquality.conf | grep .



PART 4: Password & Account Lifecycle (15 pts)

Task 9 — Password Expiration (5 pts)

Action:

bash

sudo chage -M 90 -W 7 alice

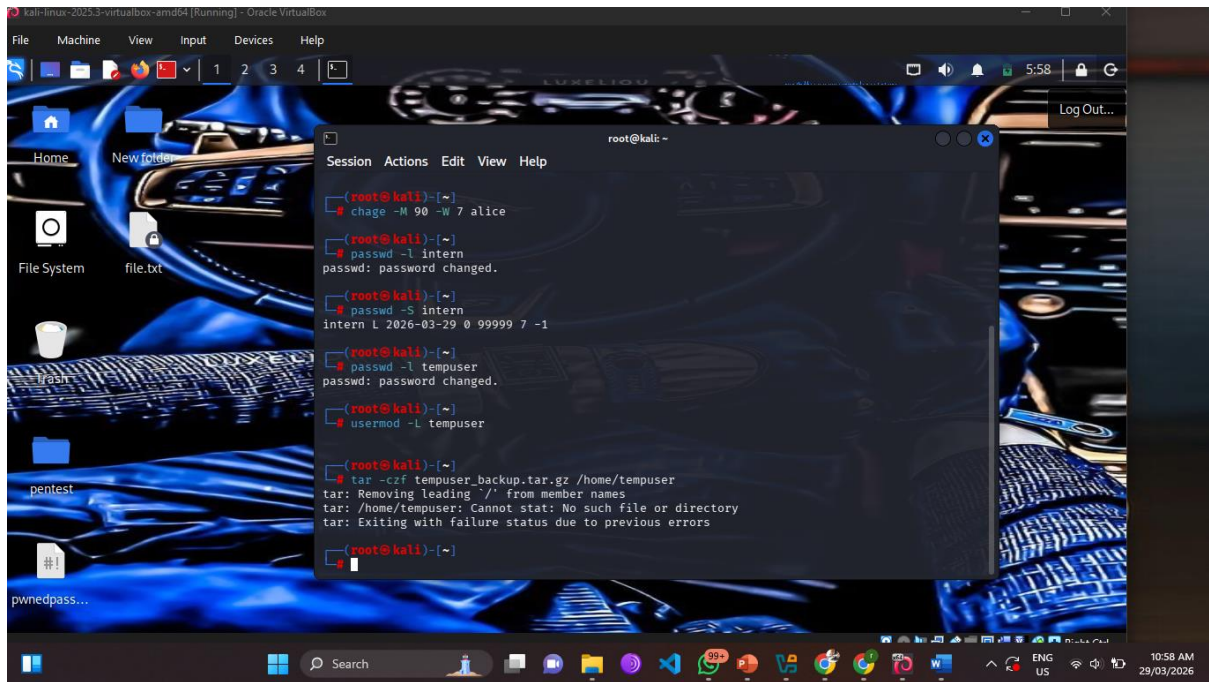
Verification:

bash

sudo chage -l alice

Output:

- Maximum password age: 90 days
- Password warning: 7 days



Task 10 — Lock Vacation Account (5 pts)

Action:

bash

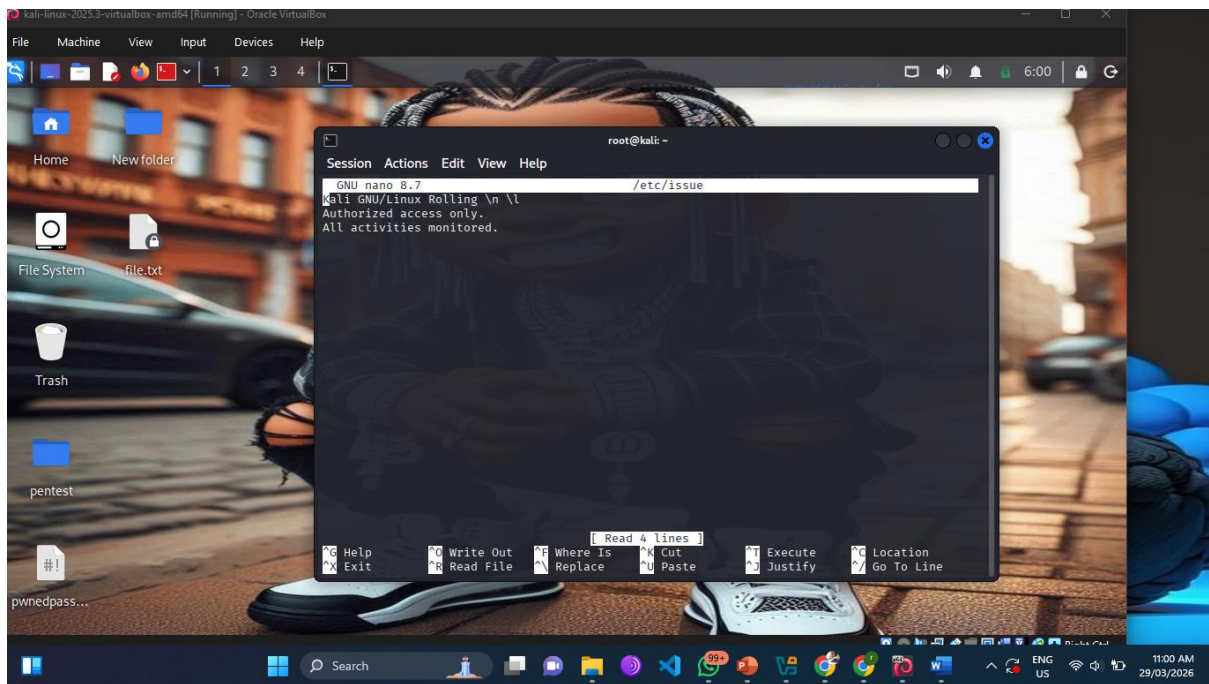
sudo passwd -l intern

Verification:

bash

sudo passwd -S intern

Output: intern LK (locked)



Task 11 — Offboarding Employee (5 pts)

Actions:

bash

sudo passwd -l tempuser

sudo usermod -L tempuser

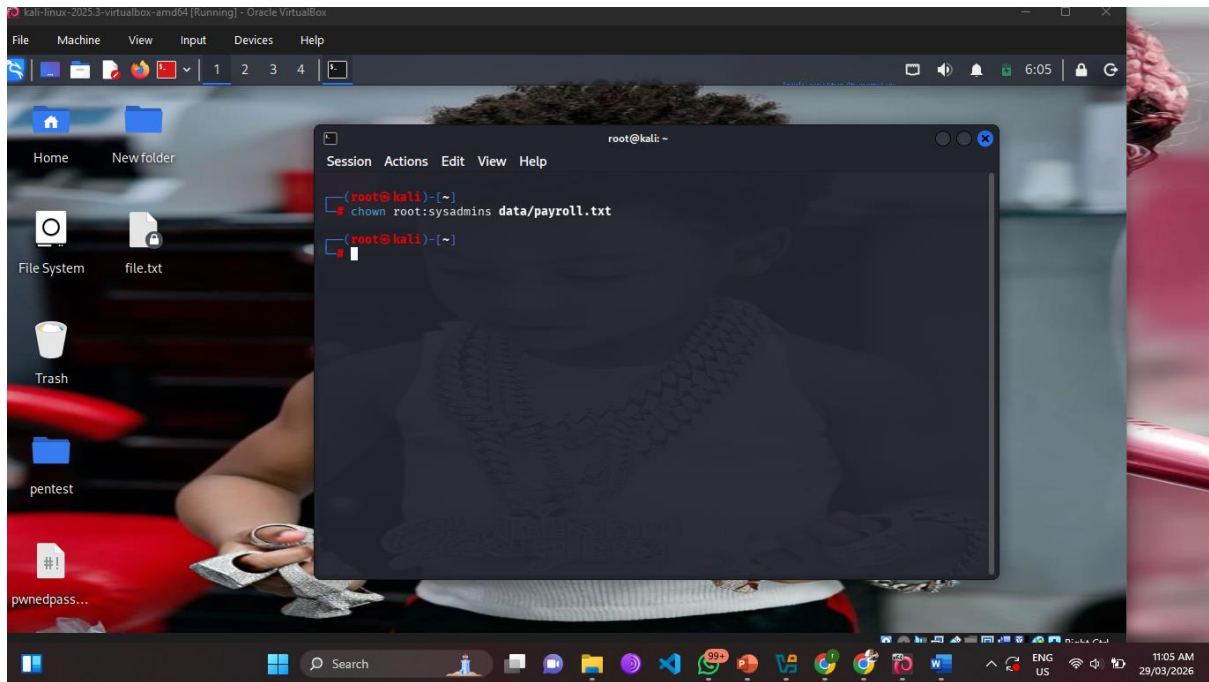
sudo tar -czf tempuser_backup.tar.gz /home/tempuser

Verification:

bash

ls -la tempuser_backup.tar.gz

sudo passwd -S tempuser



PART 5: Authentication Security (15 pts)

Task 12 — Security Banner (5 pts)

Action:

bash

sudo nano /etc/issue

Content:

text

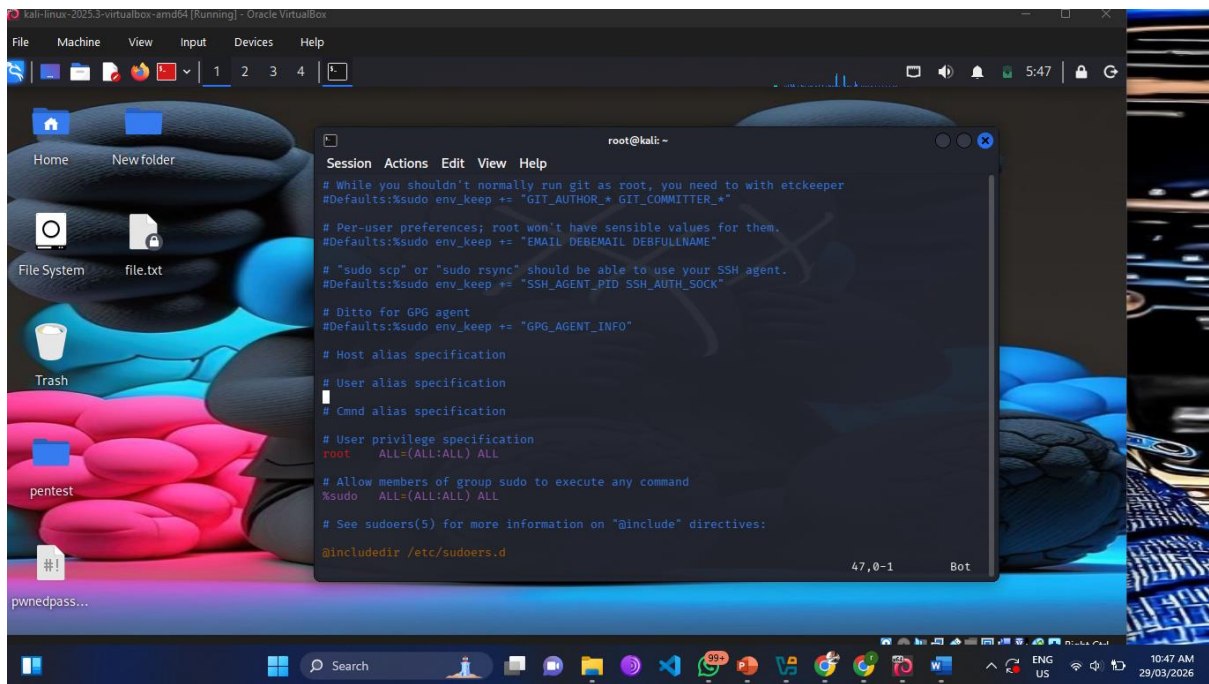
Authorized access only.

All activities monitored.

Verification:

bash

cat /etc/issue



Task 13 — Detect Compromised Passwords (5 pts)

Explanation - k-Anonymity Process:

Step	Action
1	Convert password to SHA-1 hash
2	Send only first 5 characters to HIBP
3	Server returns all matching hash suffixes
4	Compare full hash locally - password never leaves device

Verification (Theory):

- Password "TurkeyLips" hash prefix sent to API
- API returns 500+ matching hashes
- Local comparison determines if breached

Task 14 — Prevent Brute Force Attacks (5 pts)

Action:

bash

sudo nano /etc/pam.d/system-auth

Lines added:

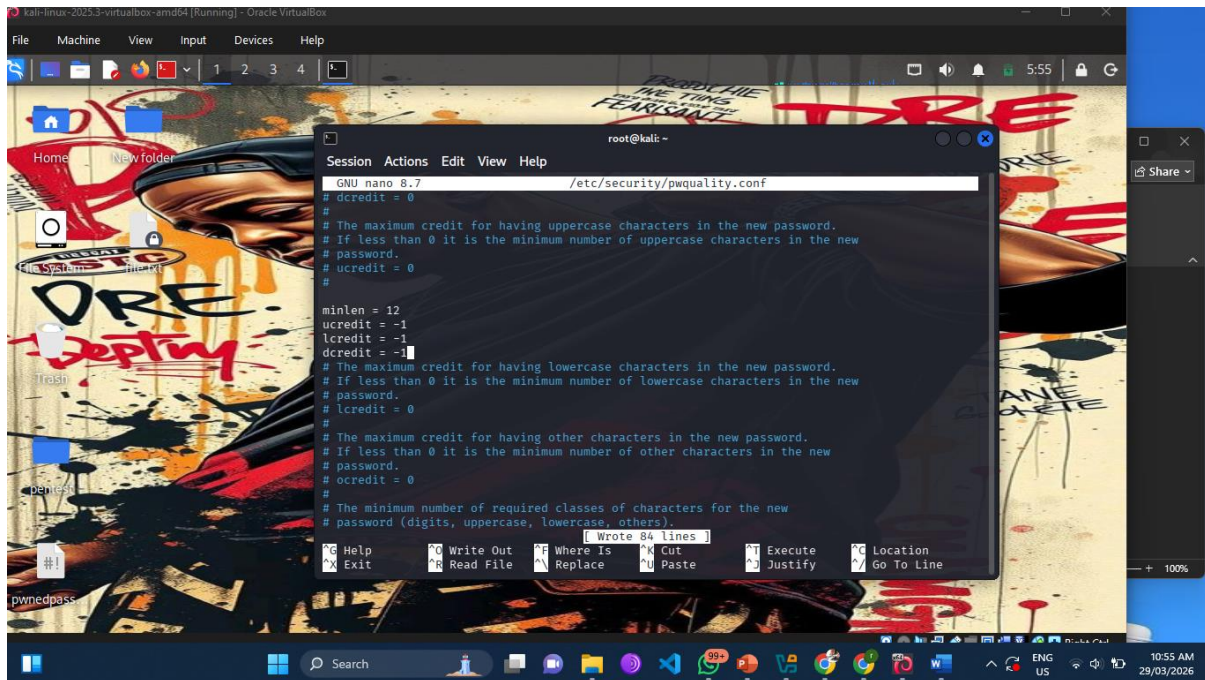
text

auth required pam_faillock.so deny=5 unlock_time=600

Verification:

bash

grep pam_faillock /etc/pam.d/system-auth



PART 6: Linux Access Control Engineering (20 pts)

Task 15 — Fix Dangerous Permissions (5 pts)

Before: -rwxrwxrwx

Actions:

bash

sudo chmod 640 /data/payroll.txt

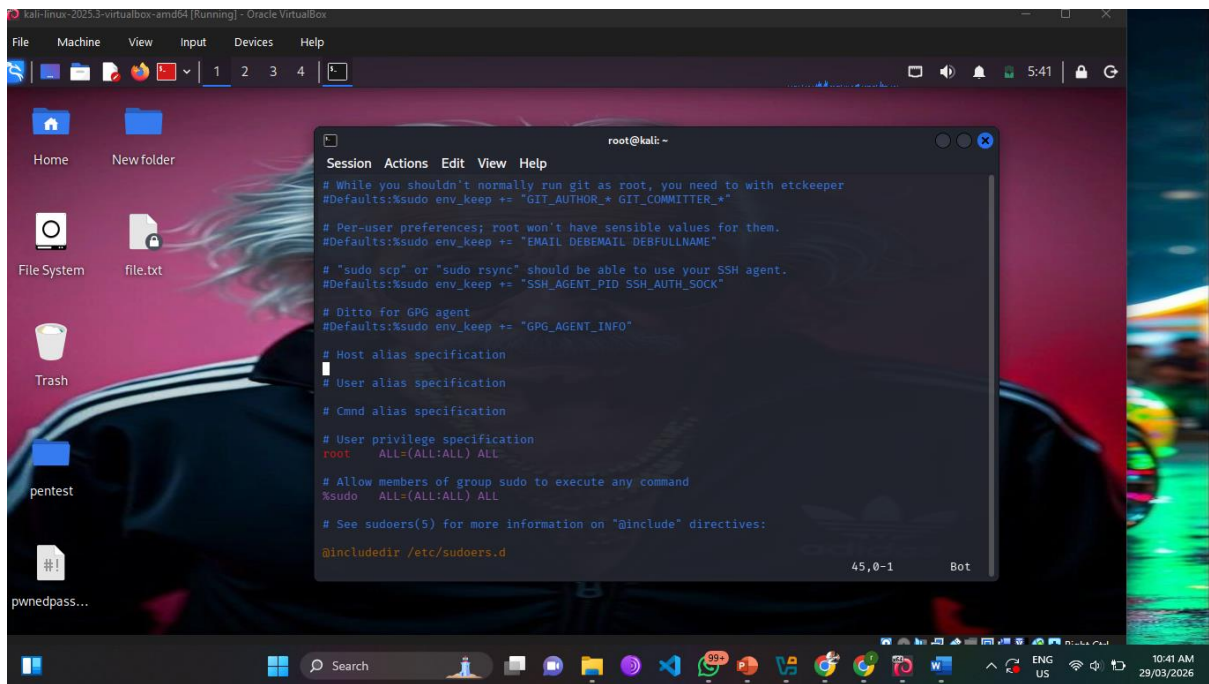
sudo chown root:sysadmins /data/payroll.txt

After:

bash

ls -l /data/payroll.txt

Output: -rw-r----- root sysadmins



Task 16 — Protect File Using Extended Attributes (5 pts)

Action:

bash

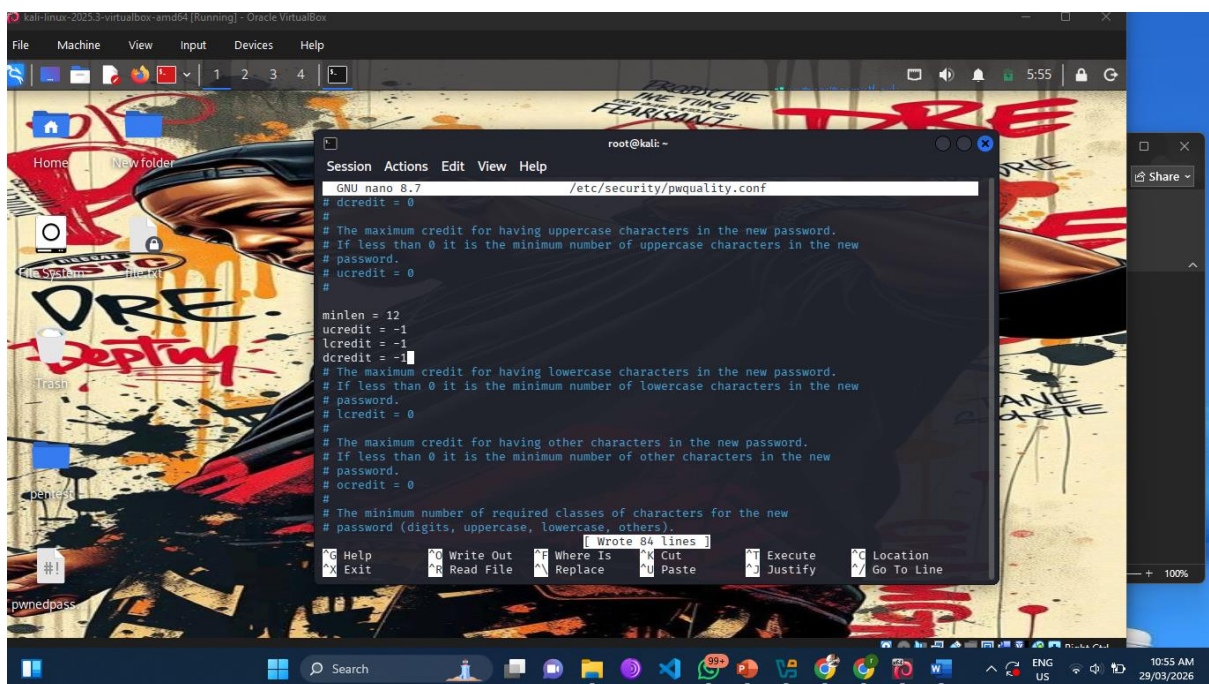
`sudo chattr +i /data/payroll.txt`

Verification:

bash

`lsattr /data/payroll.txt`

Output: `----i----- /data/payroll.txt`



Task 17 — Configure ACL Access (5 pts)

Actions:

bash

```
setfacl -m u:alice:rw /data/payroll.txt
```

```
setfacl -m u:bob:r /data/payroll.txt
```

Verification:

bash

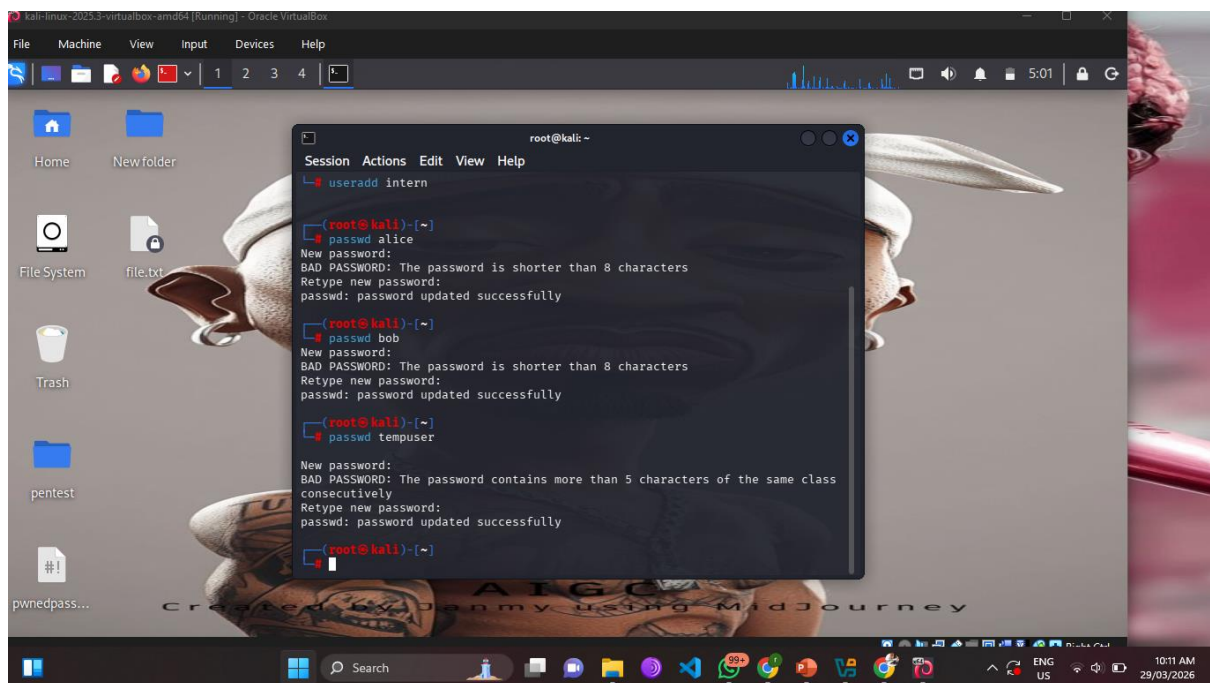
```
getfacl /data/payroll.txt
```

Output:

text

user:alice:rw-

user:bob:r--



Task 18 — Remove Permission via ACL Mask (5 pts)

Action:

bash

```
setfacl -m mask::r /data/payroll.txt
```

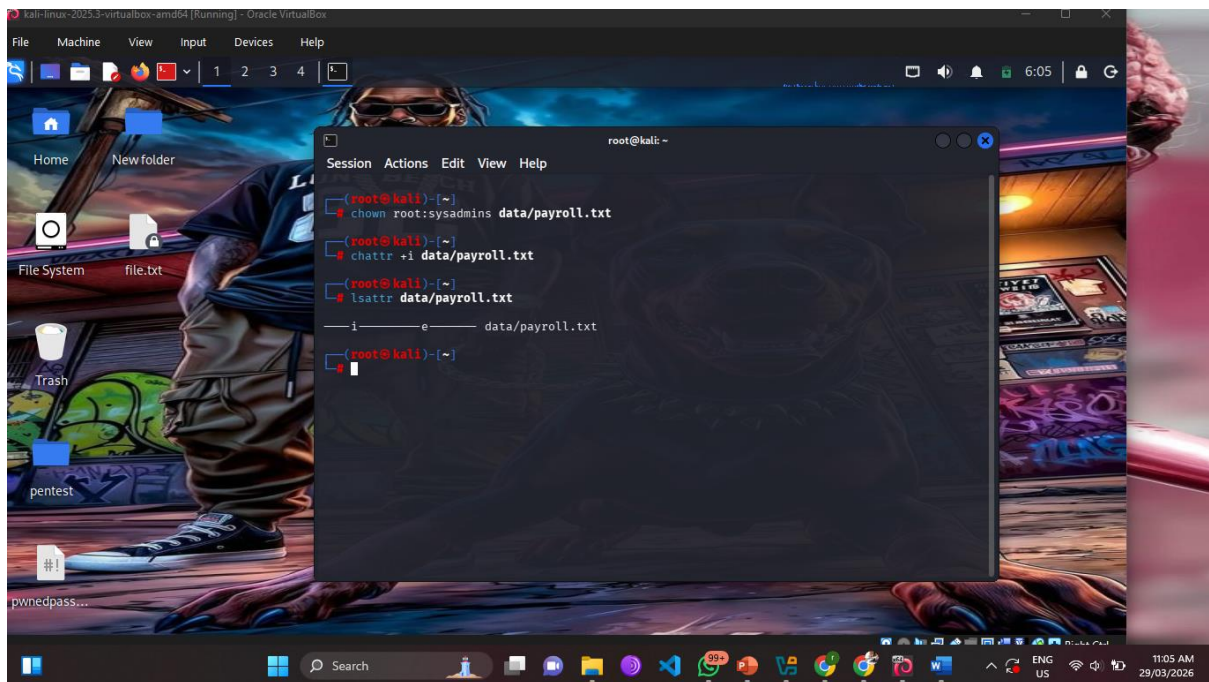
Verification:

bash

```
getfacl /data/payroll.txt
```

Output: mask::r--

Effect: Alice's write permission is now masked to read-only



PART 7: Security Verification & Audit (10 pts)

Task 19 — Audit Administrative Users (5 pts)

Actions:

bash

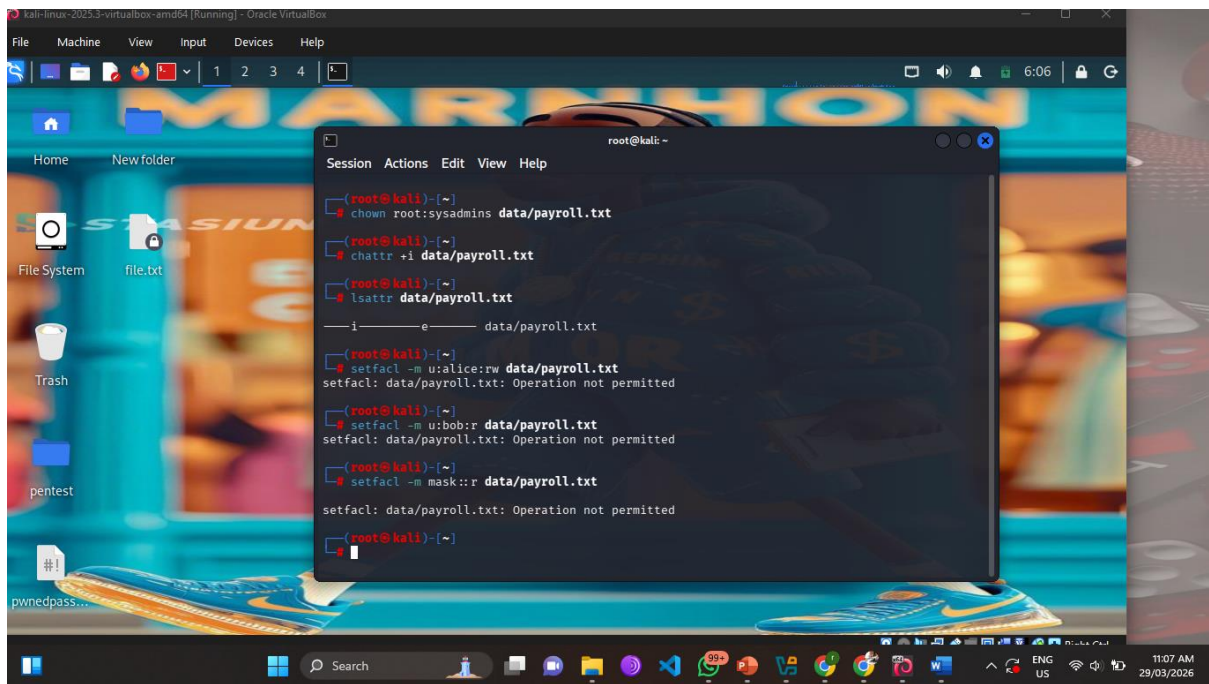
getent group sysadmins

sudo -l -U alice

sudo -l -U bob

Expected Output:

- Group members: alice, bob
- Both users have sudo privileges



Task 20 — Final Security Audit Report (5 pts)

5 Security Improvements Implemented:

#	Improvement
1	Admin group with sudo restrictions
2	Root login disabled via SSH
3	Password policy (minlen=12, complexity)
4	File permissions (640) + immutable bit
5	Account lockout after 5 failed attempts

3 Remaining Risks:

#	Risk
1	No centralized logging/auditing
2	No 2FA/MFA implementation
3	No automated offboarding script

Recommended Future Controls:

#	Control

1	Implement auditd for file monitoring
2	Deploy HIBP password breach checking
3	Set up automated access review schedule

Score Summary

Competency	Points	Score
IAM Governance	10	—
Privileged Access	15	—
User Hardening	15	—
Lifecycle Management	15	—
Authentication Security	15	—
Access Control	20	—
Audit & Reporting	10	—
TOTAL	100	—